DHANUSH KUMAR S V

PROCESS ENGINEER

Taichung, Taiwan • +886-0909505486 • dhanushkumar795@gmail.com • linkedin.com/in/dhanush-kumar-772274213 • dhanushkumarsv.github.io

SUMMARY

Process Engineer skilled in process simulation, optimization, and manufacturing improvement. Experienced in using Aspen Plus, Aspen HYSYS, and MATLAB for modeling chemical and semiconductor processes. Strong understanding of yield improvement, defect reduction, and process integration. Dedicated to enhancing production efficiency and product quality through data-driven analysis and continuous improvement.

EDUCATION

M.S. in Chemical Engineering — 2024 – Present

National Chung Hsing University — Taichung, Taiwan | GPA: 3.95 / 4.3

B.Tech in Chemical Engineering — 2019 – 2023

Anna University — Chennai, India | GPA: 7.84 / 10

TECHNICAL EXPERTISE

Process Simulation: Aspen Plus, Aspen HYSYS, MATLAB, GAMS
Process Integration: Electrochemical coating, Surface process modeling, Yield optimization
Data Analysis: Origin, Excel, ImageJ, Python
Documentation: Process flow reports, Design of experiments, Technical writing

PROJECTS

Hybrid VMD-MED / MSF-MED Process Modeling

Phosphogypsum Wastewater Treatment

- Modeled hybrid VMD-MED and MSF-MED systems for phosphogypsum wastewater purification using process simulation.
- Integrated microbial electrochemical cell (MEC) for enhanced phosphorus recovery and energy-efficient treatment.
- Performed process optimization and energy analysis to improve water recovery, phosphorus yield, and overall system efficiency.

MILP-Based Optimization of Cooperative Dairy Milk Supply Chain

Supply Chain Integration

- Formulated a GAMS-based MILP model for integrated village-BMC-plant milk procurement under capacity, flow balance, and service constraints.
- Conducted sensitivity and scenario analyses on BMC capacity and penalty parameters to evaluate cost-service trade-offs.
- Achieved ~17% total cost reduction, up to ~30% decrease in unserved milk, and identified key capacity thresholds.

Hydrogen Production using the Photocatalytic Method

Sustainable Energy Generation

- Designed and built a 4 L trapezoidal photocatalytic reactor for solar-driven hydrogen production from sulphuric wastewater.
- Achieved ~300 mL/h/L hydrogen production using TiO_2 photocatalyst under direct sunlight.
- Optimized catalyst loading and sulphide concentration to maximize hydrogen yield and process efficiency.

INTERNSHIPS & EXPERIENCE

University Teaching Assistant — Sep 2025 – Dec 2025

National Chung Hsing University, Taiwan

- Guided students in process simulation using Aspen Plus, troubleshooting modeling and convergence issues.
- Supported design and optimization of chemical processes; assisted in teaching optimization concepts using GAMS.

Quality Assurance Trainee — Oct 2022 – Nov 2022

Steel Authority of India Limited (SAIL), Salem

- Analyzed production data from hot and cold rolling lines to identify factors affecting steel yield and quality.
- Investigated manufacturing defects and contributed to root cause analysis for process improvement.

CFD Research Intern — Nov 2021 – Jan 2022

Indian Institute of Technology, Indore

- Performed computational fluid dynamics analysis for chemical processes, learning parameter optimization and reporting techniques.

Surface Coating Intern Aug 2021

RK Metals

- Studied electrochemical coating and metallization methods; evaluated coating thickness, adhesion, and corrosion resistance.

PUBLICATIONS & CONFERENCES

- "Location Selection and Purification Process Simulation for a Glycerol Plant." — ICATES 2024
- "Production of Hydrogen Gas Using Photo-catalytic Method." — ICATES 2023

LANGUAGES

Tamil (Native) • English (Fluent) • Chinese (Basic)